

Module 15 – Advanced Python Programming

1. Introduction

MediTrack is a Python-based desktop application developed for MediCare Hub to digitize patient records, appointments, billing, and user access. The application uses Object-Oriented Programming, file handling, exception handling, SQLite database connectivity, regular expressions, and a Tkinter-based GUI.

2. Patient & Appointment Management

Patient profiles are created using Python classes. Each patient object stores personal details such as patient ID, name, age, gender, and contact information. Multiple appointments and prescriptions are linked to each patient using lists or database tables. Data is stored using file operations or SQLite3 for persistent storage.

3. Doctor Assignment & Visit Tracking

Appointments are logged through a Tkinter GUI form. Doctors are assigned during appointment creation. Each visit is recorded with date, symptoms, diagnosis, and prescribed medicines. Visit history is maintained for every patient using OOP inheritance for reuse and scalability.

4. Billing & Payment History

Billing is calculated by adding consultation charges, medicine costs, and applicable tax. Exception handling is used to manage invalid or missing data. Invoices are saved in text or CSV format using file handling for future reference.

5. User Roles & Access Control

The application supports Admin, Receptionist, and Doctor roles. Login authentication ensures role-based access. Admin has full control, Receptionist manages appointments and billing, and Doctors can view patient records. Custom exceptions are used to restrict unauthorized access.

6. Regex-Based Reports

Regular expressions are implemented to search patient records by disease name or patient status. Pattern-based filtering allows users to retrieve specific cases such as all follow-up patients. This improves reporting accuracy and efficiency.

7. Key Python Concepts Used

- File Handling – read, write, and update patient records
- Exception Handling – try, except, finally, and custom exceptions
- OOP – classes, inheritance, method overriding, and use of super()
- SQLite3 – local database storage
- Tkinter – GUI forms, buttons, labels, and validation

8. Reflective Thinking

This application can be enhanced by integrating cloud-based medical records, QR code-based patient identification, and multi-branch database synchronization. These improvements will help scale MediTrack for larger healthcare networks.